

Def. Define

$$f: [0, 1] \rightarrow \mathbb{R}; x \mapsto \begin{cases} \frac{1}{p} & \text{if } x \in \mathbb{Q} \text{ with } x = \frac{q}{p}, \text{ irreducible form,} \\ 0 & \text{otherwise.} \end{cases}$$

Then,

- 1) f is continuous at every irrational numbers, and discontinuous at every rational numbers.
- 2) f is Riemann-integrable.

Proof. observe that for each $n \in \mathbb{N}$, the set

$$S_n = \{x \in [0, 1] \cap \mathbb{Q} \mid f(x) > \frac{1}{n}\} = f^{-1}\left[\left(\frac{1}{n}, \infty\right)\right] \cap \mathbb{Q}$$

is finite set. Let $x_0 \in [0, 1]$ be an irrational number.

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$. Since the S_N is finite, we can take $\delta > 0$ s.t.

$$x \in (x_0 - \delta, x_0 + \delta) \Rightarrow |f(x_0) - f(x)| = f(x) \leq \frac{1}{N} < \varepsilon.$$

so, f is continuous at any irrational. And, for rational $\frac{q}{p} \in [0, 1]$,

for any $\delta > 0$, there is an irrational $x \in (\frac{q}{p} - \delta, \frac{q}{p} + \delta)$ s.t. $|\frac{1}{p} - f(x)| = \frac{1}{p} > \frac{1}{2p}$.

so it is not continuous at rational. $\square$

Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$.

Denote $S_N = \{x_1, x_2, \dots, x_k\}$. And, for $0 < d < \min\{|x_i - x_0| \mid 1 \leq i \leq k\}$.

Now, construct a partition as

$$P = \left\{x_1 - \frac{d}{2}, x_1 + \frac{d}{2}, x_2 - \frac{d}{2}, \dots, x_k - \frac{d}{2}, x_k + \frac{d}{2}\right\} \cap [0, 1].$$

Then, for the upper sum $U(f, P)$ and $L(f, P)$,

$$U(f, P) - L(f, P) = U(f, P) < 1 \cdot \frac{1}{N} + dk < \varepsilon + dk$$

Since the k is fixed, letting $d \rightarrow 0$, we obtain the result.